PROJECT PHASE 1: PROBLEM DEFINITION AND DATA UNDERSTANDING

Deadline: Week 6
Weight: 5% of the project grade
Penalty: Each delay is penalized with **3 points**
Maximum allowed delays: 2

Objective

The goal of this phase is to clearly define the problem, understand the dataset, and position the project within existing research and real-world applications.

Coding (Data Understanding & Exploration)

In this phase, you must load and perform an initial exploration of your dataset.

Minimum requirements:

- Data loading (from source: CSV, API, database, parquet files, etc.)
- Dataset description (number of rows, columns, feature names)
- Data types analysis (numerical, categorical, text, time-series, etc.)
- Missing values analysis
- Basic statistics (mean, median, standard deviation, etc.)
- Distribution analysis (histograms, boxplots, etc.)
- At least **3–5 relevant visualizations** specific to your problem

Report

- Please use the report template provided: https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template

The report must include:

1. Problem Definition

- Clear description of the problem you aim to solve

2. Importance and Real-World Context

- Why the problem matters
- Where it is used in practice (industry/research)

3. Related Work

- At least **4–5 recent articles** (published in the last 3–4 years)

For each paper, you must include:

- Problem addressed
- Dataset used
- Methodology (models, techniques)
- Evaluation metrics
- Key results

Additionally:

- Provide a **comparison between the approaches**
- Highlight:
  - Advantages
  - Drawbacks

Deliverables (Mandatory)

- **Report:**
  - LaTeX source file
  - PDF version
- **Code:**
  - Jupyter Notebook or Python scripts
  - Must be runnable
- **Repository:**
  - All work must be committed to the **corresponding GitHub Classroom team repository**

Grading

- **Report:** 50%
- **Implementation (code & data exploration):** 50%

The final grade for Phase 1 will be the weighted average of the report and implementation. Grades will be provided at the end of the week, after evaluation by the teachers (course and laboratory).

Important

- The work **cannot be evaluated** if it is not committed to the corresponding GitHub team repository.
- Make sure your repository contains **all required files** before the deadline (before laboratory from week 6)